

Colomn Dataset tidak sesuai

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0.1 # KLASIFIKASI GENRE GAME BERDASARKAN SINOPSIS

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0.2 Library & Load Dataset

library

```
[1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import MultiLabelBinarizer

from sklearn.naive_bayes import MultinomialNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import LinearSVC

from sklearn.metrics import classification_report, confusion_matrix, \
    accuracy_score, f1_score, precision_score, recall_score

import re
import langdetect
from langdetect import detect
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from nltk.tokenize import word_tokenize
import nltk
```

```
[2]: # Jika pertama kali saja:
"""
nltk.download('punkt')
nltk.download('stopwords')
"""
```

```
[2]: "\nnltk.download('punkt')\nnltk.download('stopwords')\n"
```

0.3 Load dan Memeriksa Data (Data Loading & EDA)

Load Data

```
[6]: df = pd.read_csv('games.csv', encoding='utf-8', low_memory=False)
df.head()
```

```
[6]:
```

	AppID	Name	Release date	\
2539430	Black Dragon Mage Playtest	Aug 1, 2023	0 - 0	
496350	Supipara - Chapter 1 Spring Has Come!	Jul 29, 2016	0 - 20000	
1034400	Mystery Solitaire The Black Raven	May 6, 2019	0 - 20000	
3292190	- Vtuber Paranoia	Oct 31, 2024	0 - 20000	
3631080	Maze Quest VR	Apr 24, 2025	0 - 20000	

	Estimated owners	Peak CCU	Required age	Price	Discount	DLC count	\
2539430	0	0	0.00	0		0	
496350	0	0	5.24	65		0	
1034400	0	0	4.99	0		0	
3292190	1	0	8.99	0		1	
3631080	0	0	4.99	0		0	

	About the game	\
2539430	NaN	
496350	Springtime, April: when the cherry trees come ...	
1034400	Immerse yourself in the most beloved, mystical...	
3292190	synopsis 'Hello, I'm Hiyoro, a new YouTuber!' ...	
3631080	Its not just a Maze; its a Quest! Enter the ca...	

	Supported languages	...	\
2539430	[]	...	
496350	['English']	...	
1034400	['English', 'French', 'German', 'Russian']	...	
3292190	['Korean']	...	
3631080	['English']	...	

	Average playtime two weeks	Median playtime forever	\
2539430	0	0	
496350	0	8	
1034400	0	0	
3292190	0	0	
3631080	0	0	

	Median playtime two weeks	Developers	Publishers	\
2539430	0	NaN	NaN	
496350	0	minori	MangaGamer	
1034400	0	Somer Games	8floor	

3292190	0
3631080	0 Reality Expanded LLC Reality Expanded LLC

	Categories \
2539430	NaN
496350	Single-player,Steam Trading Cards,Steam Cloud,...
1034400	Single-player,Family Sharing
3292190	Single-player,Steam Achievements,Family Sharing
3631080	Single-player,VR Only,Steam Leaderboards,Famil...

	Genres \
2539430	NaN
496350	Adventure
1034400	Casual
3292190	Casual,Indie,Simulation
3631080	Action,Early Access

	Tags \
2539430	NaN
496350	Adventure,Visual Novel,Anime,Cute
1034400	Casual,Card Game,Solitaire,Puzzle,Hidden Objec...
3292190	NaN
3631080	NaN

	Screenshots	Movies
2539430	https://shared.akamai.steamstatic.com/store_it...	NaN
496350	https://shared.akamai.steamstatic.com/store_it...	NaN
1034400	https://shared.akamai.steamstatic.com/store_it...	NaN
3292190	https://shared.akamai.steamstatic.com/store_it...	NaN
3631080	https://shared.akamai.steamstatic.com/store_it...	NaN

[5 rows x 39 columns]

Informasi Dasar Dataset

```
[4]: print("Informasi Dataset:")
df.info()
```

Informasi Dataset:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 122611 entries, 0 to 122610

Data columns (total 39 columns):

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	AppID	122611 non-null	int64
1	Name	122610 non-null	object
2	Release date	122611 non-null	object
3	Estimated owners	122611 non-null	object

4	Peak CCU	122611	non-null	int64
5	Required age	122611	non-null	int64
6	Price	122611	non-null	float64
7	DiscountDLC count	122611	non-null	int64
8	About the game	122611	non-null	int64
9	Supported languages	114162	non-null	object
10	Full audio languages	122611	non-null	object
11	Reviews	122611	non-null	object
12	Header image	12070	non-null	object
13	Website	122530	non-null	object
14	Support url	49676	non-null	object
15	Support email	54142	non-null	object
16	Windows	100348	non-null	object
17	Mac	122611	non-null	bool
18	Linux	122611	non-null	bool
19	Metacritic score	122611	non-null	bool
20	Metacritic url	122611	non-null	int64
21	User score	4256	non-null	object
22	Positive	122611	non-null	int64
23	Negative	122611	non-null	int64
24	Score rank	122611	non-null	int64
25	Achievements	40	non-null	float64
26	Recommendations	122611	non-null	int64
27	Notes	122611	non-null	int64
28	Average playtime forever	22458	non-null	object
29	Average playtime two weeks	122611	non-null	int64
30	Median playtime forever	122611	non-null	int64
31	Median playtime two weeks	122611	non-null	int64
32	Developers	122611	non-null	int64
33	Publishers	114174	non-null	object
34	Categories	113702	non-null	object
35	Genres	113658	non-null	object
36	Tags	114198	non-null	object
37	Screenshots	83346	non-null	object
38	Movies	116593	non-null	object

dtypes: bool(3), float64(2), int64(15), object(19)

memory usage: 34.0+ MB

Pilih kolom yang relevan

```
[5]: # Seleksi kolom yang relevan

df = df[["Name", "About the game", "Genres", "Supported languages"]]

# Ganti nama kolom agar konsisten
df = df.rename(columns={
    "About the game": "synopsis",
    "Genres": "genres",
```

```

    "Supported languages": "languages"
})

df.head()

```

```

[5]:
      Name synopsis \
0      Black Dragon Mage Playtest      0
1  Supipara - Chapter 1 Spring Has Come!      0
2      Mystery Solitaire The Black Raven      0
3          - Vtuber Paranoia      1
4          Maze Quest VR      0

      genres \
0      NaN
1  Single-player,Steam Trading Cards,Steam Cloud,...
2      Single-player,Family Sharing
3  Single-player,Steam Achievements,Family Sharing
4  Single-player,VR Only,Steam Leaderboards,Famil...

      languages
0      NaN
1  Springtime, April: when the cherry trees come ...
2  Immerse yourself in the most beloved, mystical...
3  synopsis 'Hello, I'm Hiyoro, a new YouTuber!' ...
4  Its not just a Maze; its a Quest! Enter the ca...

```

0.4 PRE-Prosessing

0.5 Modeling & Training

0.6 Evaluasi Model

0.7 Implementasi